

1. Administrative & Study Identification

Full Study Title: A Phase II Multi-center, Single Arm, Safety and Efficacy Study of MBG453 in Combination With Azacitidine and Venetoclax for the Treatment of Acute Myeloid Leukemia (AML) in Adult Patients Unfit for Chemotherapy **Short Title:** A Study of MBG453 in Combination With Azacitidine and Venetoclax in AML Patients Unfit for Chemotherapy (STIMULUS-AML1) **Protocol Number:** CMBG453C12201 **NCT Number:** NCT04150029 **Posting ID:** 874617068820700877 **Sponsor/Company:** Novartis **Investigational Product:** Sabatolimab (MBG453) + Azacitidine + Venetoclax **Phase of Development:** Phase 2 **Trial Status:** TERMINATED (Study terminated early) **Medical Monitor:** Novartis Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: Part 1 (Safety Run-in) aimed to rule out excessive toxicity of MBG453 when administered with azacitidine and venetoclax. The combined Part 1 and Part 2 aimed to evaluate the safety, tolerability, and efficacy (specifically Complete Remission [CR] rate) of combining MBG453 with azacitidine and venetoclax in adult patients with newly diagnosed Acute Myeloid Leukemia (AML) who are unfit for standard intensive chemotherapy. **Secondary Objectives:** To determine Measurable Residual Disease (MRD) Negativity Rate and Duration of Complete Remission (CR).

2.2 Background & Rationale To address the critical unmet medical need for patients with newly diagnosed AML who are ineligible for intensive chemotherapy. While the combination of azacitidine and venetoclax (AZA+VEN) has improved outcomes in this population, patients still face high relapse rates and short overall survival. Sabatolimab (MBG453) is an investigational immuno-myeloid therapy that binds to TIM-3. This trial sought to extend preliminary efficacy findings by evaluating MBG453 in combination with hypomethylating agents (HMA) and the BCL-2 inhibitor venetoclax.

Note: Novartis decided to end the development of this compound. Consequently, the planned primary analysis (initially planned after all subjects completed at least 12 cycles) was skipped, and a final study analysis was performed instead. At the final analysis timepoint, only 1 patient had not completed the 12 cycles of treatment or discontinued earlier.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Targeted Enrollment (\$N\$): 90 participants **Number of Sites:** International, multi-center **Study Status:** TERMINATED (Development of compound ended by Novartis)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult patients (≥ 18 years of age at the date of signing the ICF). 2. Newly diagnosed with AML based on 2016 WHO classification. 3. Unfit and not suitable for intensive chemotherapy defined by: age ≥ 75 , ECOG performance status 2 or 3, or specific severe comorbidities: severe cardiac comorbidities (including congestive heart failure, LVEF $\leq 50\%$, chronic stable Angina), pulmonary comorbidity (DLCO $\leq 65\%$ or FEV1 $\leq 65\%$), moderate hepatic impairment (total Bilirubin >1.5 to $3\times$ ULN), renal impairment (eGFR ≥ 30 to <45 mL/min/ $1.73m^2$), or other comorbidity incompatible with intensive chemotherapy. 4. Not planned for hematopoietic stem-cell transplantation (HSCT). 5. Eastern Cooperative Oncology Group (ECOG) performance status of 0, 1, 2, or 3.

4.2 Exclusion Criteria 1. Prior exposure to TIM-3 directed therapy or history of severe hypersensitivity reactions to any ingredient of study drugs (azacitidine, venetoclax, or MBG453) or monoclonal antibodies (mAbs) and/or their excipients. 2. Previous treatment at any time with checkpoint inhibitors, venetoclax, or hypomethylating agents (HMAs) such as decitabine or azacitidine. 3. Current use or use within 14 days prior to randomization of systemic steroid therapy (> 10 mg/day prednisone or equivalent) or any immunosuppressive therapy. (Topical, inhaled, nasal, ophthalmic, replacement therapy, or steroids for transfusion are allowed). 4. Active autoimmune disease requiring systemic therapy (e.g., corticosteroids). 5. Live vaccine administered within 30 days prior to randomization.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Sabatolimab (MBG453) administered in combination with standard cycles of venetoclax (daily) and azacitidine (7 days per cycle) over 28-day treatment cycles. **Route of Administration:** Sabatolimab (Intravenous), Azacitidine (Intravenous/Subcutaneous), Venetoclax (Oral). **Duration of Treatment:** Continued until disease progression, unacceptable toxicity, or patient withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Topical, inhaled, nasal, ophthalmic steroids; replacement steroid therapy; and

steroids given in the context of a blood transfusion. **Prohibited:** Systemic immunosuppressive therapy (> 10 mg/day prednisone equivalent), any other investigational or approved antineoplastic agents, and live vaccines administered within 30 days prior to randomization.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Comorbidity Assessment, Baseline Bone Marrow Biopsy/Aspiration
Baseline	Day 0	First Dose, Safety Run-in Labs, Vital Signs
Interim	Every 28 Days (Cycle Day 1)	Response Assessment (Bone Marrow/Labs), Efficacy Assessment, Safety and DLT Monitoring
End of Study	Follow-up	Final Vital Signs, Exit Interview, Disease Progression Logging, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes

- 1. Incidence of Dose Limiting Toxicities (DLT) (Safety run-in Patients Only):** Defined as an adverse event or abnormal laboratory value considered at least possibly related to MBG453 as a single contributor or in combination that occurs during the DLT observation period.
- 2. Percentage of Participants Achieving Complete Remission (CR Rate):** As per investigator assessment based on IWG (Cheson et al 2003) and ELN 2017 (Dohner et al 2017).

7.2 Secondary Outcomes

- 1. Measurable Residual Disease (MRD) - Negativity Rate:** Full Study Population.
- 2. Measurable Residual Disease (MRD) - Negativity Rate:** In participants with Best Overall Response (BOR) of CR/CRi and evaluable MRD.
- 3. Duration of Complete Remission (CR).**

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard tracking for immune-mediated adverse events and hematologic toxicities, including febrile neutropenia, thrombocytopenia, and anemia.

Serious Adverse Events (SAEs): Reported to the sponsor and regulatory bodies within standard 24-hour timelines.

Data Monitoring Committee (DMC): Utilized a Safety Review Committee during the initial dose-escalation/safety run-in phase to rule out excessive toxicity of the combination.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Efficacy analysis performed on all evaluable patients with newly diagnosed AML; Safety set includes all patients receiving ≥ 1 dose of any study drug. **Sample**

Size Justification: Targeted enrollment of 90 patients, sized appropriately to rule out excessive toxicity of the triplet combination and evaluate a preliminary efficacy signal.

Termination Impact on Analysis: Originally, a primary analysis of the CR rate was planned after all subjects completed at least 12 cycles (28 days each) or discontinued. Due to the termination of the compound's development by Novartis, this primary analysis was skipped in favor of a final study analysis.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Continuous safety monitoring, onsite monitoring visits, and central data reviews throughout the treatment cycles.

Audit Trail: Detailed electronic Case Report Forms (eCRFs) ensuring accountability of administration for the three-drug regimen.

11. Study Identifiers & Governance

Primary ID: CMBG453C12201 **Registry Number:** NCT04150029 **Posting**

ID: 874617068820700877 **Sponsor/Lead Organization:** Novartis **Study**

Title: A Study of MBG453 in Combination With Azacitidine and Venetoclax in AML Patients Unfit for Chemotherapy **Official**

Regulatory Title: A Phase II Multi-center, Single Arm, Safety and Efficacy Study of MBG453 in Combination With Azacitidine and Venetoclax for the Treatment of Acute Myeloid Leukemia (AML) in Adult Patients Unfit for Chemotherapy

12. Clinical Framework (AML Specific)

Primary Condition: Acute Myeloid Leukemia (AML) **Diagnosis**

Threshold: Newly diagnosed per 2016 WHO classification; ineligible for standard intensive chemotherapy. **Medical**

Methodology: Triplet systemic therapy: Immuno-myeloid (anti-TIM-3 / MBG453) + Hypomethylating agent (Azacitidine) + BCL-2 inhibitor (Venetoclax). **Optimization Target:** Achievement of

Complete Remission (CR) and targeted elimination of leukemic stem cells (LSCs).

13. Study Procedures & Methodology

Management Pathway: Frontline systemic management for intensive chemotherapy-ineligible AML. **Education Protocol (HCP):** Training on proper dosing schedules for the combination therapy and management algorithms for immune-mediated or hematologic adverse events. **Education Protocol (Patient):** Guidance on recognizing signs of neutropenia/infection and adherence to the oral venetoclax regimen. **Quality Audit Mechanism:** Local and centralized validation of bone marrow pathology results and DLT evaluations.

14. Observation & Follow-Up Schedule

Total Duration: Study terminated early; protocol designed for treatment continuing until progression or unacceptable toxicity. **Onsite Visit Cadence:** Preceding every 28-day treatment cycle (Cycle Day 1). **Remote Monitoring Frequency:** Between treatment cycles as required for symptom tracking. **Checklist Review Interval:** Disease assessment via bone marrow biopsy per standard protocol guidelines (e.g., end of Cycle 1, and subsequently as clinically indicated).

15. Sign-off & Approval

Role	Name	Signature	Date	---	---	---	---
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]				
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]				